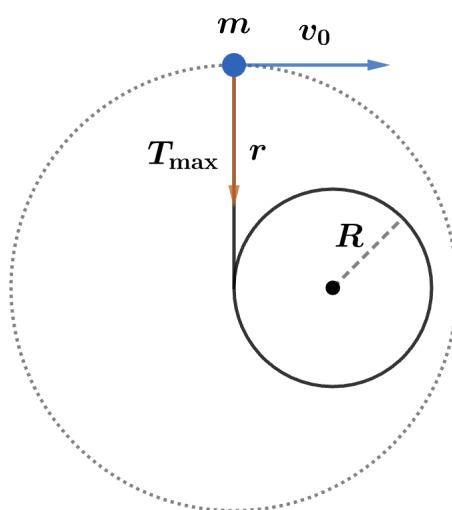


2007 F=ma Exam: Problem 36

Kevin S. Huang



We found (in problem 34) that the length of the unwound rope when it breaks is

$$r = \frac{mv_0^2}{T_{\text{max}}}$$

so the answer is D.